# Introduction to Labour Market Economics

EC306- Wages and Employment

Justin Smith

Wilfrid Laurier University

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# Goals of This Section

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In this section we will:

- Define **Labour Economics**
- Describe basic facts about the labour market
- Introduce the main economic model used in labour economics
- Discuss some current policy issues

# What is Labour Economics?

# Defining Labour Economics

Key Definition

**Labour Economics** is the study of the interaction between workers, employers, and government that determine outcomes like wages and employment

Key aspects of labour economics:

- Labour supply: decisions by individuals about how much to work
- Labour demand: decisions by firms about how many workers to hire
- Labour market: the interaction of labour supply and demand that determines wages and employment
- Other determinants of wages: human capital, discrimination, unions, minimum wage, etc.
- Inequality and poverty: the distribution of income and wealth
- Unemployment: causes and consequences

## Think About It

If labour economics studies the interaction between workers, employers, and government, what would happen to wages and employment if one of these three players was removed entirely?

# Labour Supply

- Studies decisions by individuals about how much to work
- There are many key determinants
  - Wages: higher wages increase the opportunity cost of leisure, leading to more work
    - Can also lead to *less* work if wages are already very high
  - Non-labour income: higher non-labour income reduces the need to work, leading to less work
  - Preferences: some people may prefer more leisure or more work
  - Public policy: taxes, transfers, and regulations can affect labour supply
    - Examples: income tax, welfare programs, minimum wage, pension, workers compensation

# Labour Demand

- Studies decision by firms about how many workers to hire
- Key determinants
  - Wages: higher wages increase the cost of labour, leading to less hiring
  - Other labour costs: benefits, training, and other costs increase the cost of labour, leading to less hiring
  - Productivity: more productive workers are more valuable, leading to more hiring
  - Output prices: higher output prices increase revenue, leading to more hiring
  - Technology: new technology can change the demand for different types of workers
  - Public policy: taxes, subsidies, and regulations can affect labour demand
    - Examples: corporate tax, employment subsidies, minimum wage, occupational licensing

# Relative Wages

- Across the economy, different groups of workers earn different wages
  - Men earn more than women
  - Education is positively related to wages
  - More dangerous jobs tend to pay more
  - Public sector jobs pay differently than private sector ones
  - Immigrants earn less than non-immigrants
- Labour Economics uses supply, demand and other models to explain these differences
  - Returns to schooling explained by **human capital theory**
  - Gender and immigrant wage gaps partly explained by discrimination

# Unions

- Unions are organizations that represent workers in collective bargaining with employers
- Unions increase worker power, and can affect wages and employment
- Typically wages of unionized workers are determined by **collective bargaining**
  - Leads to a collective bargaining agreement, renegotiated every few years
- To study unionized workers, labour economists incorporate bargaining theory into their models
- Unions can also affect non-wage aspects of jobs, such as benefits, working conditions, and job security

# Unemployment

- Most topics in labour economics use **microeconomic** models to explain behaviour
- Unemployment is studied using both microeconomic and macroeconomic models

Typical microeconomic topics include:

- Job search: how workers find jobs and how firms find workers
- Implicit contracts: agreements between workers and firms that are not formal contracts
- Efficiency wages: wages that are above the market-clearing level to increase productivity and reduce turnover
- Employment insurance: government programs that provide income support to unemployed workers

# Unemployment

Typical macroeconomic topics include:

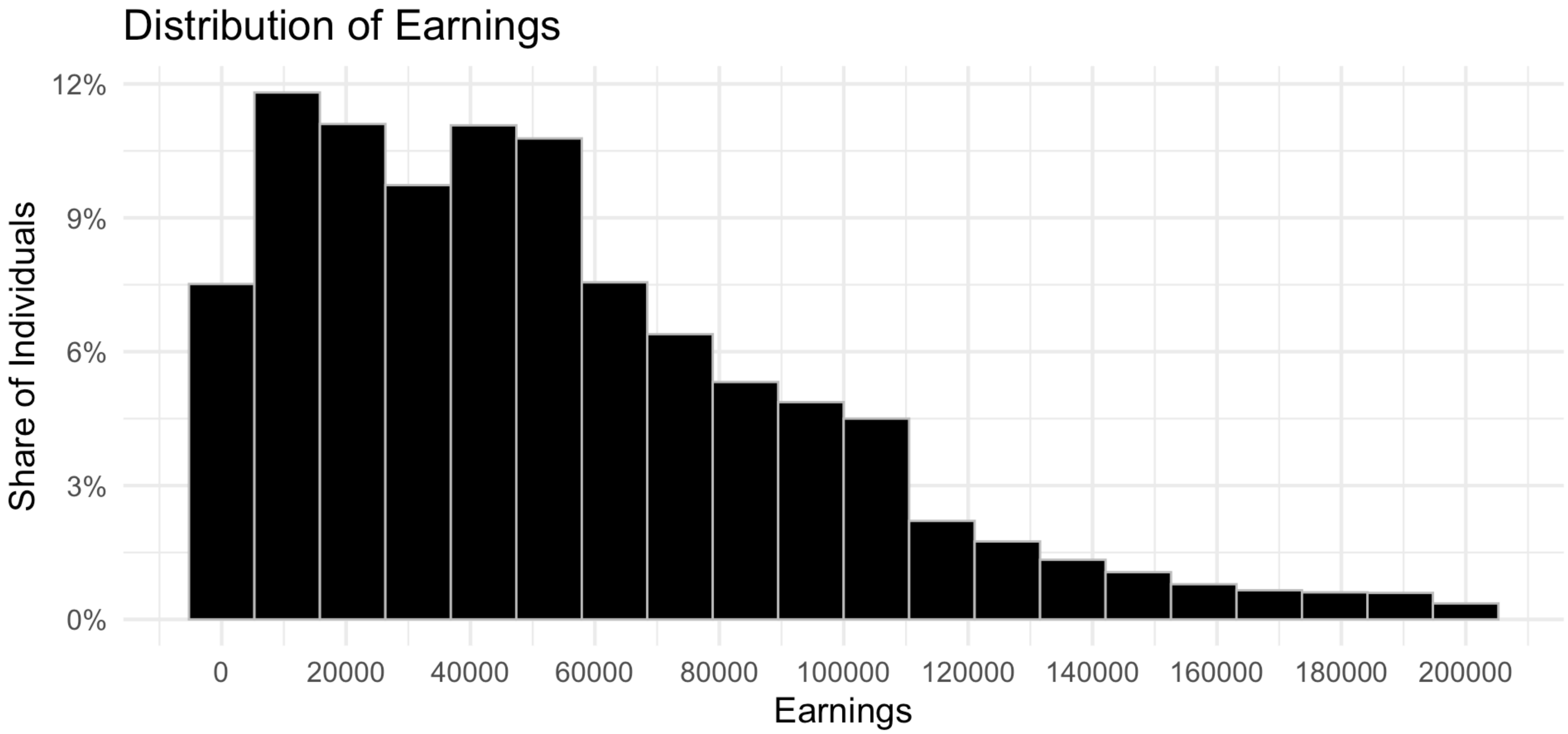
- Business cycles: fluctuations in economic activity that affect employment
- Inflation: the general increase in prices that can affect real wages and employment
- Productivity: the amount of output produced per worker that can affect wages and employment

# Statistics on Labour Market Outcomes

# Income Sources

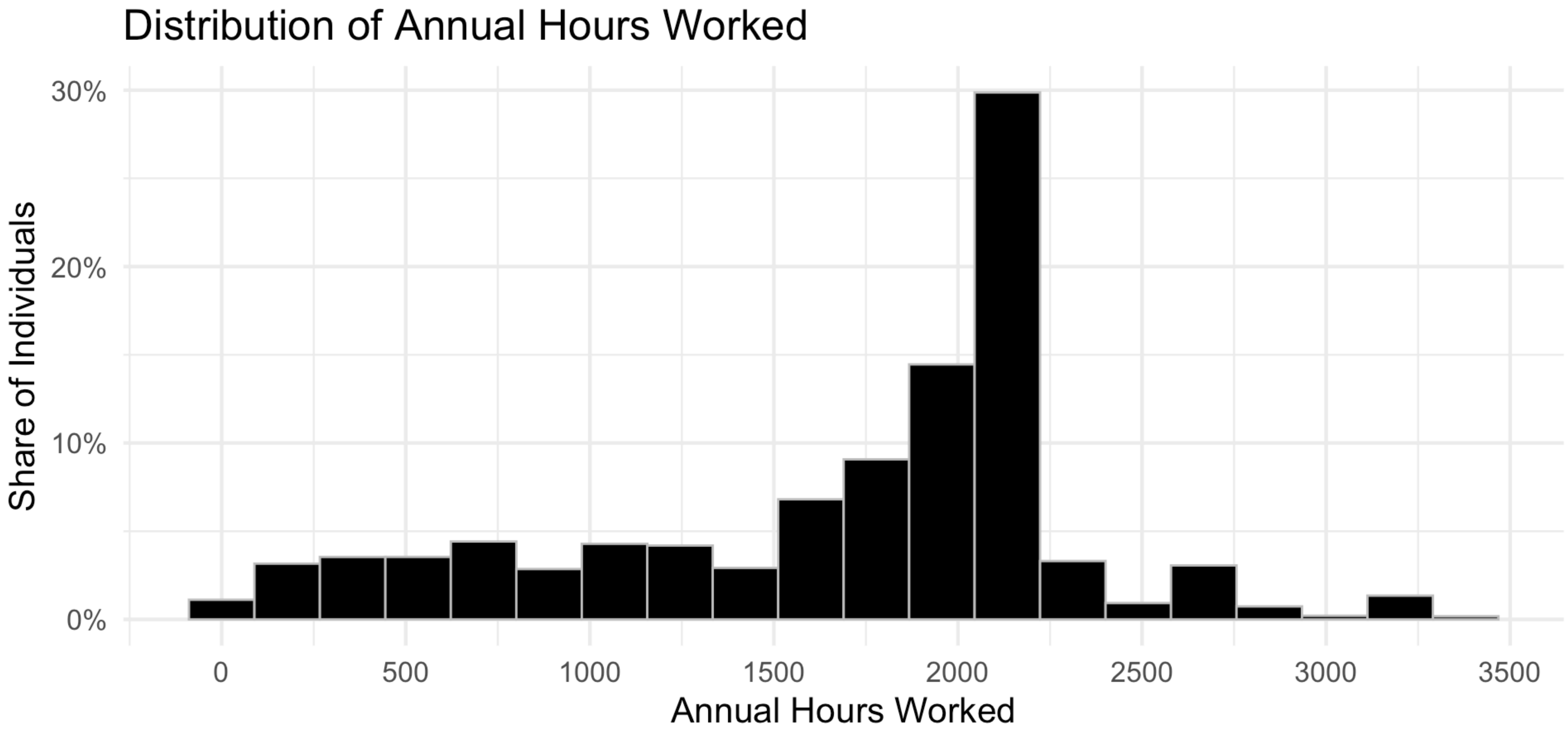
| | All | | Men | | Women | |
|---|---|---|---|---|---|---|
| **Variable** | **Mean** | **% Positive** | **Mean** | **% Positive** | **Mean** | **% Positive** |
| Earnings | 50,326 | 84.0 | 60,216 | 86.6 | 40,408 | 81.5 |
| Investment Income | 2,706 | 28.5 | 3,291 | 28.6 | 2,120 | 28.3 |
| Pension | 1,561 | 8.2 | 1,556 | 7.3 | 1,567 | 9.1 |
| Other Income | 2,069 | 17.7 | 1,926 | 16.0 | 2,212 | 19.4 |
| Government Transfers | 4,671 | 82.6 | 3,252 | 78.0 | 6,094 | 87.2 |
| Income Taxes | 11,012 | 72.8 | 14,191 | 77.2 | 7,824 | 68.5 |
| Annual Hours Worked | 1,385 | 82.0 | 1,546 | 85.6 | 1,223 | 78.4 |
| Wage | 40 | 75.6 | 44 | 78.4 | 36 | 72.9 |

# Distribution of Earnings



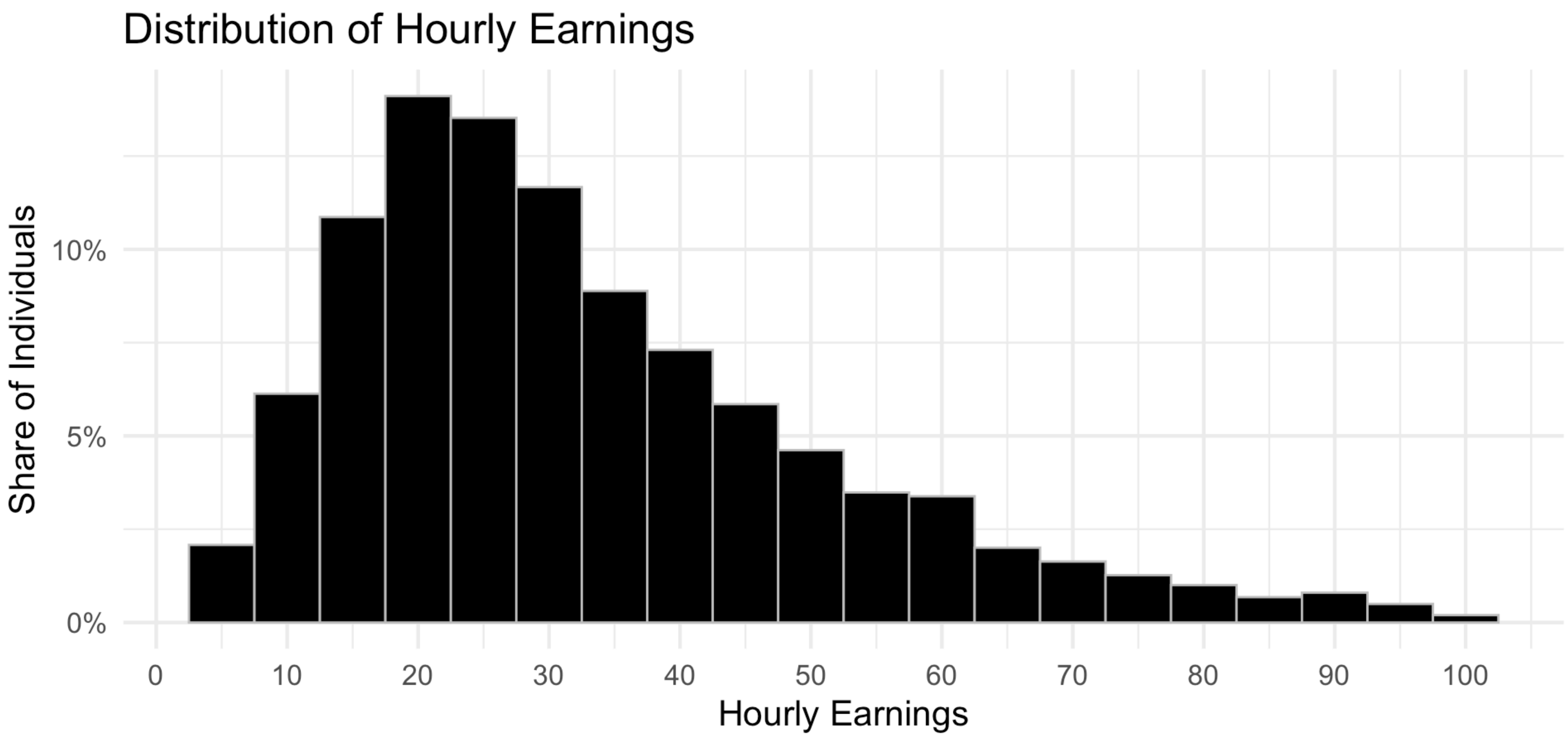

*Source: Canadian Income Survey (2022). Plot includes earnings from the 1st percentile to the 98th percentile*

# Distribution of Hours Worked



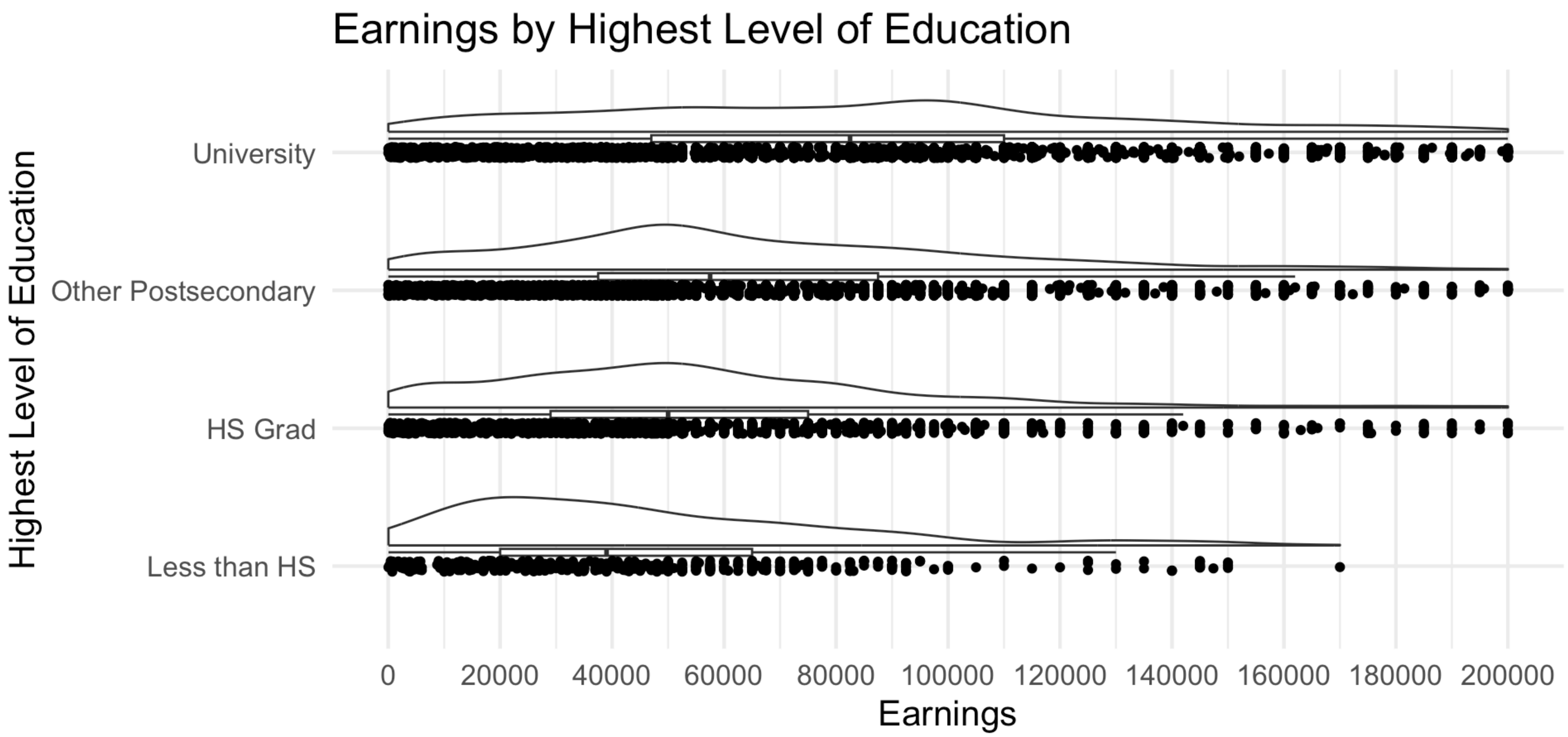

*Source: Canadian Income Survey (2022). Plot includes hours from the 1st percentile to the 99th percentile*

# Distribution of Hourly Earnings



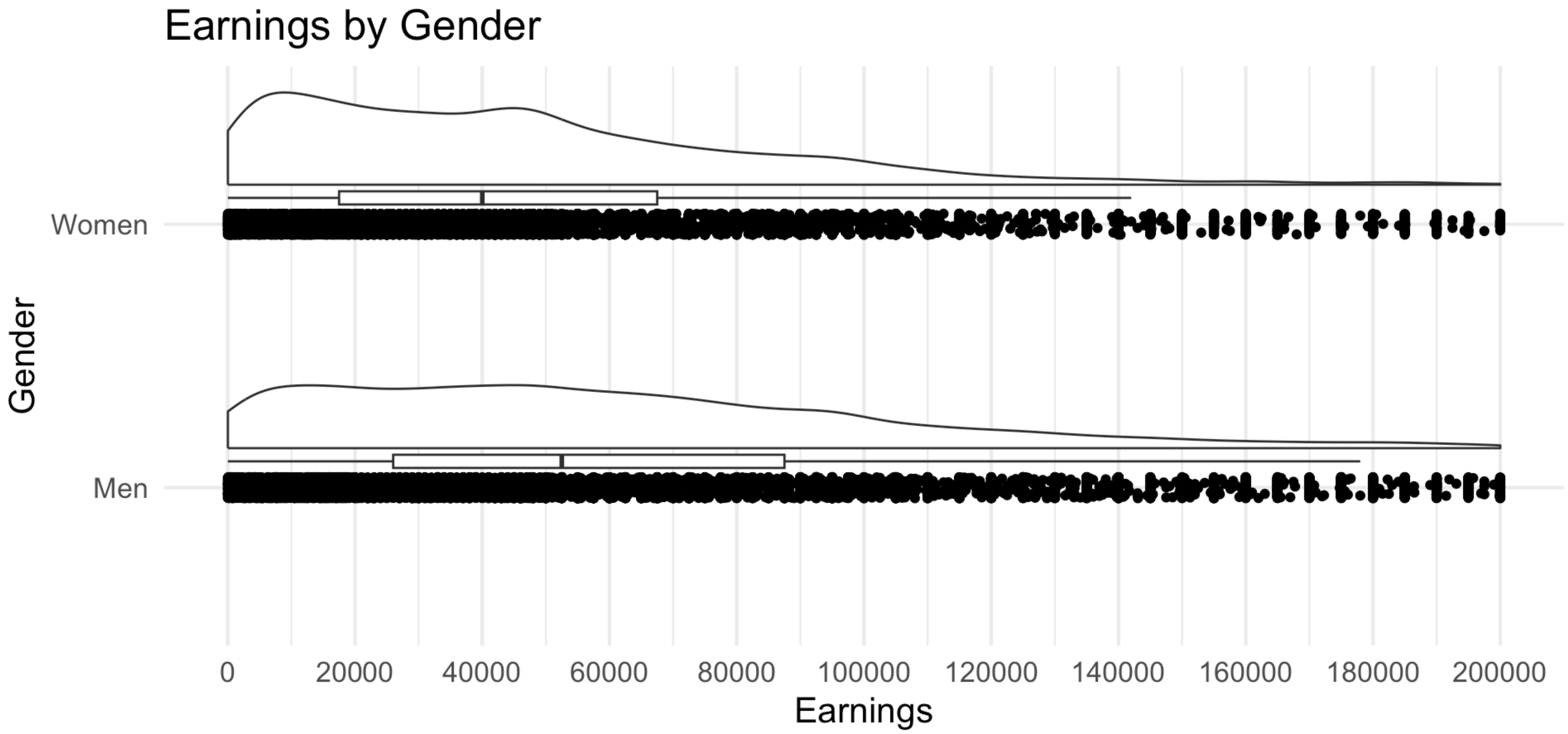

*Source: Canadian Income Survey (2022). Plot includes wages from the $5 to $100*

# Earnings by Highest Level of Schooling



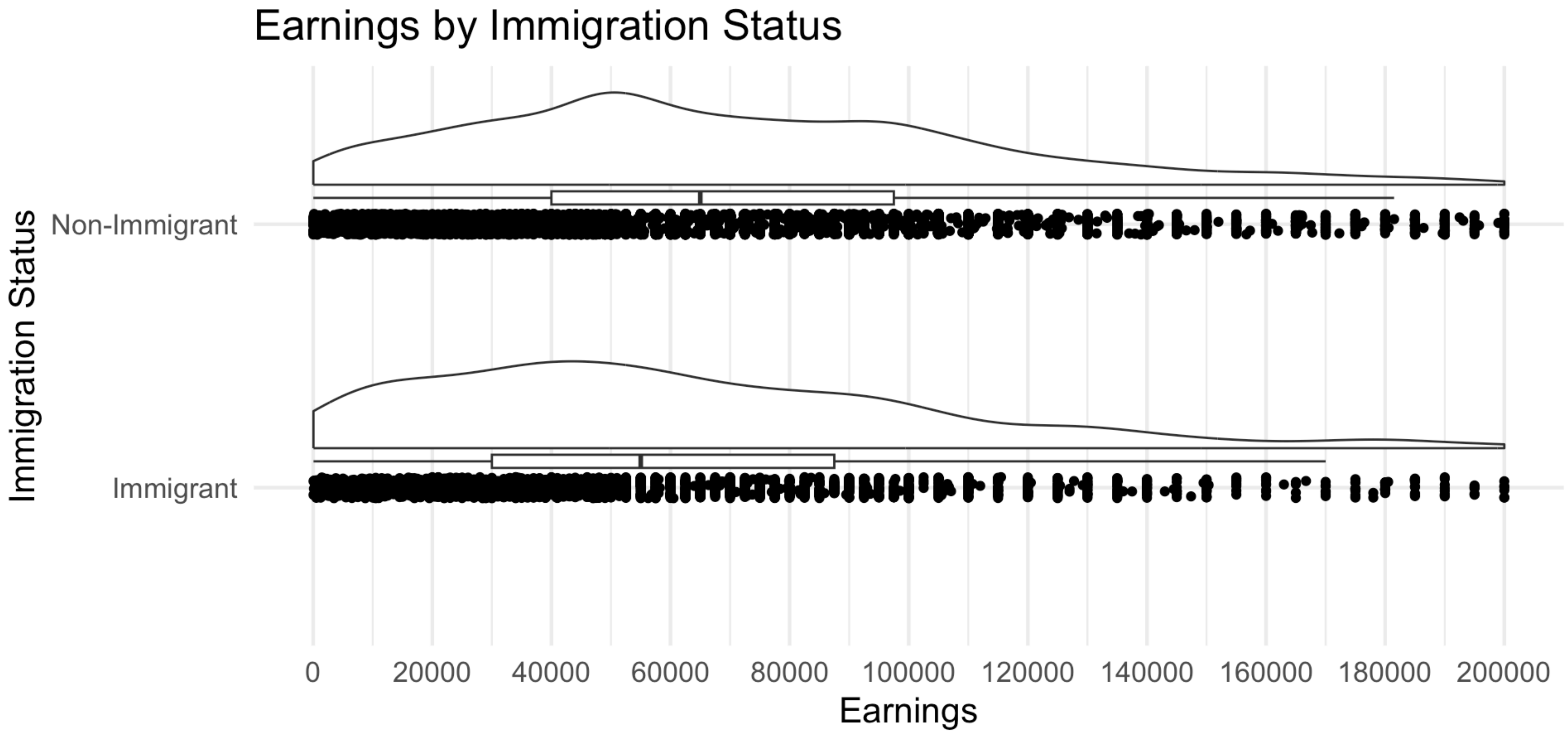

*Source: Canadian Income Survey (2022). Plot includes individuals aged 40-50, earnings from the 1st percentile to the 98th percentile*

# Earnings by Gender




*Source: Canadian Income Survey (2022). Plot includes earnings from the 1st percentile to the 98th percentile*

# Earnings by Immigration Status




*Source: Canadian Income Survey (2022). Plot includes aged 40-50, earnings from the 1st percentile to the 98th percentile*

## Discussion

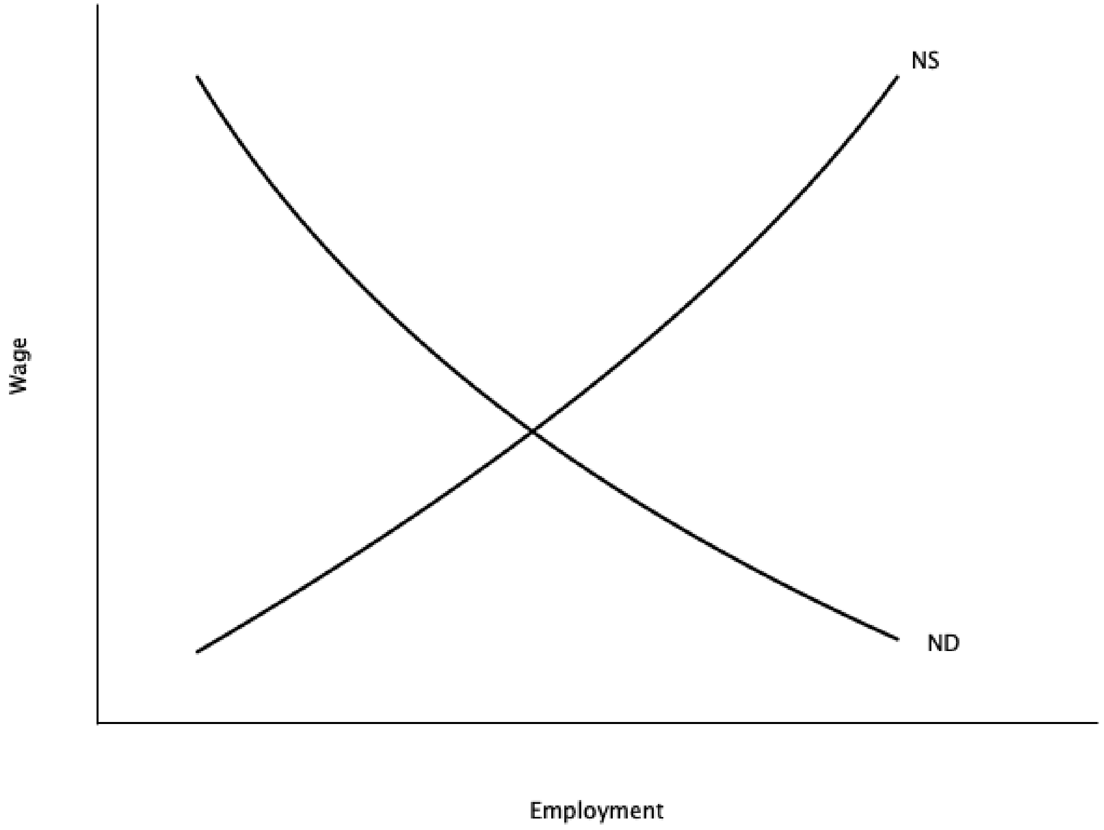
Why might we observe that men earn more than women on average, even in the same occupation? What are some possible explanations — both economic and non-economic?

# Basic Economic Model of the Labour Market

# Introduction

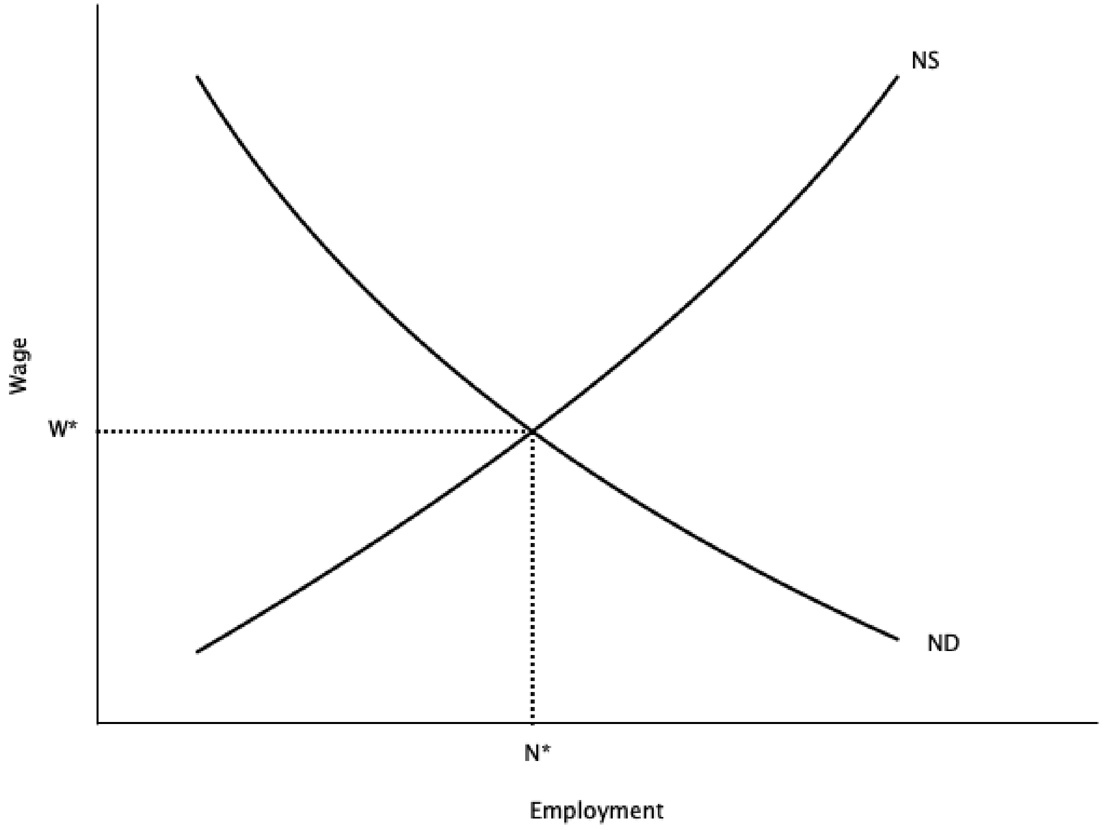
- The basic economic model of the labour market is based on **supply and demand**
- The labour market is an **input market**
  - Firms demand labour to produce goods and services
  - Workers supply labour in exchange for wages
  - Different from **output markets** where firms supply goods and services, and consumers demand them
- The interaction of labour supply and labour demand determines the equilibrium wage and employment level
- We will cover the basics here, and fill in details through the course

# Competitive Labour Market




- Graph shows demand curve (ND) and supply curve (NS) for labour
- Supply curve is upward sloping
  - Shows how much labour people will sell at each wage **all else equal**
  - The higher the wage, the higher the quantity of labour supplied
- Demand curve is downward sloping
  - Shows how much labour a firm wants to hire at each wage **all else equal**
  - The higher the wage, the lower the quantity of labour demanded

# Competitive Labour Market




- Equilibrium is where demand and supply intersect
  - Wage is W* and level of employment is N*
- Why is this equilibrium?
  - If W > W*, excess supply of labour
    - Unemployed workers willing to work for less
    - Puts downward pressure on wage
  - If W < W*, excess demand for labour
    - Firms cannot find enough workers
    - Puts upward pressure on wage

# Competitive Labour Market

- There are some key considerations of this model

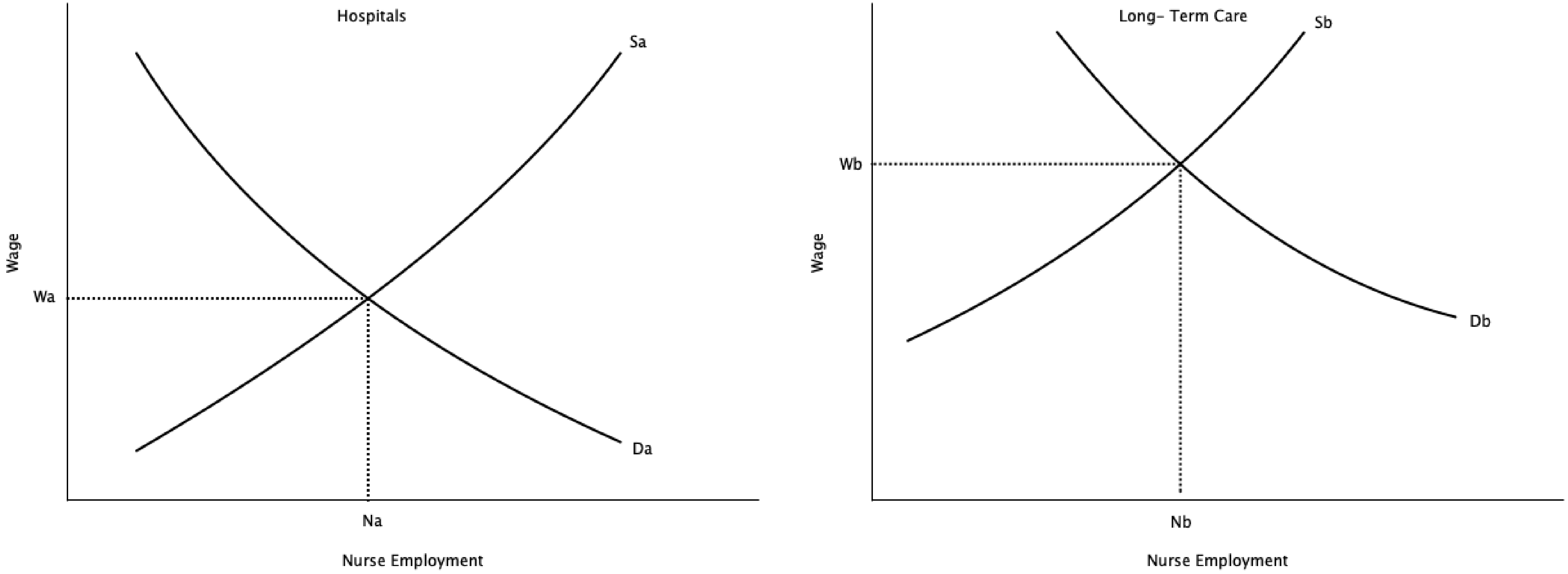
**Key Assumptions**

1. Assumes many buyers and sellers
    - No single firm or worker can influence the wage
    - Need a different model for labour markets with a single buyer (monopsony)
2. Implies no **involuntary unemployment**
    - Everyone who wants to work at the equilibrium wage can find a job
    - People who are unemployed are choosing to be
    - Not very realistic because we know the involuntary unemployed exist
3. Assumes homogeneous set of workers
    - Same in terms of their productive characteristics

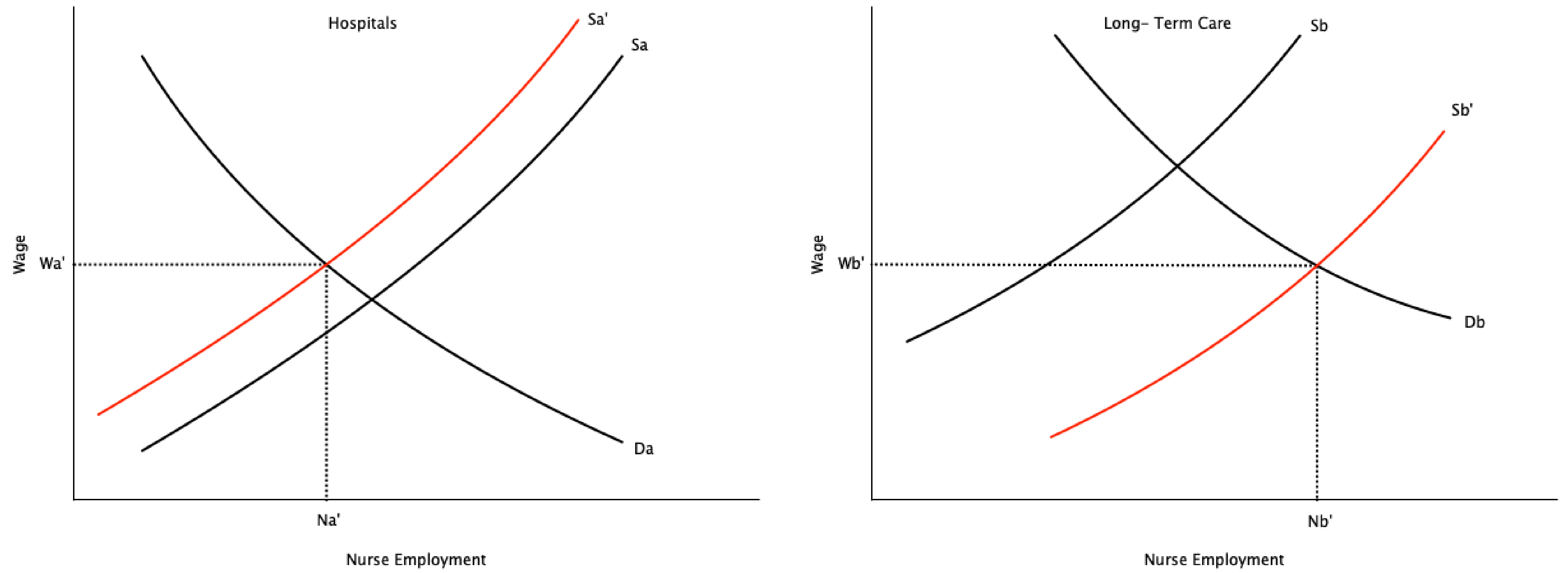
# Competitive Labour Market

- An implication of homogeneous labour is that different sectors that hire the same labour pay them the same
- Example: Nurses
  - Nurses in hospitals and long-term care facilities have similar skills and training
  - If hospitals pay more, nurses will leave long-term care to work in hospitals
  - This will continue until wages are equalized
- Other examples
  - Teachers in public and private schools
  - Electricians in construction and manufacturing
  - Data scientists in tech firms and banks

# Competitive Labour Market



# Competitive Labour Market



# Competitive Labour Market

- Graphs show a temporary imbalance in pay between hospitals and long-term care in the market for nurses
- Right graph shows long term care paying more
- Over time nurses move from hospitals to long term care
  - Movement stops when wages are equalized
- Employment in long term care increases, employment in hospitals decreases

> **Warning**
>
> This depends on key assumptions:
>
> - No barriers to moving between sectors
> - No differences in non-wage aspects of jobs
> - Homogeneous labour

- We will explore situations in the course where these assumptions do not hold

## Think About It

What would happen to wages and employment in long-term care facilities if the government made it difficult for nurses to switch between the hospital and long-term care sectors?

# Current Policy Issues

# Introduction

- Most researchers in labour economics analyze various policy issues
  - Use data to estimate their effects
  - Some do experiments
  - Others (not many) create theoretical models
- There are many policies that affect workers and firms with respect to labour
  - In Canada they can be at the federal, provincial, or municipal level
  - Can be policies from other countries (e.g. U.S. tariffs)
- Below we outline some current issues
  - Helpful for the future if you want to write a 481 paper in this area
- Textbook has an extremely long list; we will cover a few here

# Issues in Labour Supply

- How does income tax affect employment?
- Do income support programs (e.g. welfare, Canada Worker Benefit) affect employment?
- How has the Labour Force Participation Rate of women changed over time?
- How do child care costs affect employment of mothers?
- How do pension programs affect retirement decisions?
- Do temporary foreign workers affect the wages of domestic workers?

# Issues in Labour Demand

- How do minimum wages affect employment?
- Do unions affect wages and employment?
- Does technological change (e.g. Automation, AI) reduce employment?
- Does uncertainty (e.g. Trump tariffs, Brexit) affect hiring/firing decisions?
- Do employment subsidies (e.g. Canada Summer Jobs program) increase employment?

# Issues in Wage Structures

- Why do men and women earn different wages?
- Why do immigrants earn lower wages on entry?
- Do immigrant wages catch up after being in Canada for a while?
- What is the return to additional schooling?
- Do public sector workers earn more than private sector workers?
- Do workers in more dangerous jobs earn more?
- Why are CEOs paid so much?

# Issues in Unemployment

- What are the main causes of unemployment?
- How do unemployment insurance programs affect unemployment?
- How quickly does unemployment fall after a recession?
- How long do people typically stay unemployed?
- Does long term unemployment hurt future job prospects?

# Discussion

Pick one of the policy issues listed above. How would you design a study to measure its effect on the labour market? What data would you need?